Functional Erdős–Turán Universality for Comparable-Rate Luce Permutations

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Working branch — functional and multivariate strengthening

Abstract

Let σ_n be the Luce permutation induced by independent exponential clocks whose rates, after a common rescaling, lie in $[1,\kappa]$ for one fixed $\kappa<\infty$. This working branch strengthens the scalar Erdős–Turán theorem to a process-level universality statement. If $C_{n,r}$ is the number of r-cycles, set

$$K_n(t)=\sum_{r\le n^t} C_{n,r}, \qquad \mathcal{O}_n(t)=\log \operatorname{lcm}\{r\le n^t : C_{n,r}>0\}, \qquad 0\le t\le 1.$$

We prove, conditional only on the switching inputs already isolated in the scalar paper, that

$$\left(\frac{K_n(t)-t\log n}{\sqrt{\log n}}, \frac{\mathcal{O}_n(t)-\frac{1}{2}t^2(\log n)^2}{\sqrt{\frac{1}{3}(\log n)^3}}\right)_{0\le t\le 1} \Rightarrow \left(W(t), \sqrt{3}\int_0^t u\, dW(u)\right)_{0\le t\le 1}$$

in $D([0,1],\mathbb{R}^2)$. The main new intermediate statement is a high-dimensional product-Poisson approximation: throughout the growing window

$$(\log n)^3 < r \le \frac{n}{(\log n)^{A_0}},$$

the entire cycle-count vector is asymptotically independent with coordinates $\mathrm{Pois}(1/r)$ in total variation. It follows from an approximate Campbell identity obtained directly from the marked one-switch formula and a configuration-space Poisson Stein argument. The existing three-clock and von Mangoldt estimates then transfer the additive process to the truncated least-common-multiple process uniformly in time. We also derive an isonormal Gaussian limit for bounded logarithmic-scale cycle weights and the joint endpoint law

$$\left(\frac{K_n-\log n}{\sqrt{\log n}}, \frac{\log \operatorname{ord}(\sigma_n)-\frac{1}{2}(\log n)^2}{\sqrt{\frac{1}{3}(\log n)^3}}\right) \Rightarrow N\left(0, \begin{pmatrix} 1 & \sqrt{3}/2 \\ \sqrt{3}/2 & 1 \end{pmatrix}\right).$$

The final section records how this qualitative process proof interfaces with, but does not replace, the frequency-sensitive Berry–Esseen branch.

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1 Introduction and theorem map

For a permutation $\pi \in S_n$,

$$\operatorname{ord}(\pi) = \operatorname{lcm}\{\text{cycle lengths of } \pi\}.$$

For a uniform permutation, Erdős and Turán proved that $\log \operatorname{ord}(\pi)$ is asymptotically Gaussian after centering by $\frac{1}{2}(\log n)^2$ and scaling by $\sqrt{\frac{1}{3}(\log n)^3}$ [6]. The scalar comparable-rate Luce theorem proves that this normalization survives arbitrary bounded triangular arrays of clock rates; see [9]. The purpose of the present branch is to identify the full logarithmic-scale Gaussian field behind that endpoint law.

Throughout, W denotes a standard Brownian motion. Define

$$K_n(t) = \sum_{r \le n^t} C_{n,r}, \qquad B_n(t) = \sum_{r \le n^t} (\log r) C_{n,r}, \tag{1}$$

and

$$\mathcal{O}_n(t) = \log \operatorname{lcm}\{r \le n^t : C_{n,r} > 0\}. \tag{2}$$

The convention is that the least common multiple of the empty set is one.

Theorem 1.1 (Bivariate functional Erdős–Turán theorem)**.** *Let σ_n be a Luce permutation whose positive rates satisfy*

$$\sup_n \frac{\max_i \theta_{n,i}}{\min_i \theta_{n,i}} < \infty.$$

Then, in $D([0,1],\mathbb{R}^2)$ with the J_1 topology,

$$\boxed{\left(\frac{K_n(t)-t\log n}{\sqrt{\log n}},\frac{\mathcal{O}_n(t)-\frac{1}{2}t^2(\log n)^2}{\sqrt{\frac{1}{3}(\log n)^3}}\right)_{0\le t\le 1}\Rightarrow\left(W(t),\sqrt{3}\int_0^t u\;dW(u)\right)_{0\le t\le 1}.} \tag{3}$$

The convergence is uniform over all triangular arrays satisfying a fixed comparability bound.

The limiting covariance kernels are

$$\operatorname{Cov}(W(s),W(t))=s\wedge t, \tag{4}$$

$$\operatorname{Cov}\left(W(s),\sqrt{3}\int_0^t u\;dW(u)\right)=\frac{\sqrt{3}}{2}(s\wedge t)^2, \tag{5}$$

$$\operatorname{Cov}\left(\sqrt{3}\int_0^s u\;dW(u),\sqrt{3}\int_0^t u\;dW(u)\right)=(s\wedge t)^3. \tag{6}$$

In particular, the second coordinate alone has the same law as $B(t^3)$ for another standard Brownian motion B.

Corollary 1.2 (Joint endpoint law)**.** *Writing $K_n=K_n(1)$,*

$$\boxed{\left(\frac{K_n-\log n}{\sqrt{\log n}},\frac{\log \operatorname{ord}(\sigma_n)-\frac{1}{2}(\log n)^2}{\sqrt{\frac{1}{3}(\log n)^3}}\right)\Rightarrow N\left(0,\begin{pmatrix}1&\sqrt{3}/2\\ \sqrt{3}/2&1\end{pmatrix}\right).} \tag{7}$$

The process theorem follows from a stronger intermediate-window statement.

Theorem 1.3 (Product-Poisson approximation of the intermediate cycle field)**.** *After the normalization $1\le\theta_{n,i}\le\kappa$, choose the cutoffs b_n,U_n in (25). Let*

$$\xi_n=(C_{n,r})_{b_n<r\le U_n}.$$

Let $(Z_{n,r})_{b_n<r\le U_n}$ be independent with

$$Z_{n,r}\sim\operatorname{Pois}(1/r).$$

Then

$$\boxed{d_{\mathrm{TV}}\left(\mathcal{L}(\xi_n),\bigotimes_{b_n<r\le U_n}\operatorname{Pois}(1/r)\right)\longrightarrow 0.} \tag{8}$$

The convergence is uniform over all rate arrays in $[1,\kappa]$.

This theorem is stronger than what is required for a scalar characteristic function. It says that the entire growing cycle vector in the range that carries the Erdős–Turán variance is asymptotically the classical independent Poisson field.

There is also a convenient Gaussian-random-measure formulation. Put $\ell_n=\log n$ and $u_{n,r}=\log r/\ell_n$.

Theorem 1.4 (Logarithmic-scale Gaussian cycle field)**.** *For every fixed collection of bounded Riemann-integrable functions $f_1, \dots, f_d : [0,1] \to \mathbb{R}$, define*

$$\mathbb{G}_n(f) = \frac{1}{\sqrt{\ell_n}} \left[\sum_{r=1}^{n} f(u_{n,r}) C_{n,r} - \sum_{r=1}^{n} \frac{f(u_{n,r})}{r} \right]. \tag{9}$$

Then

$$\big(\mathbb{G}_n(f_1), \dots, \mathbb{G}_n(f_d)\big) \Rightarrow N(0, \Sigma), \qquad \Sigma_{ij} = \int_0^1 f_i(u) f_j(u)\, du. \tag{10}$$

Equivalently,

$$\mathbb{G}_n(f) \Rightarrow \int_0^1 f(u)\, dW(u)$$

in every fixed finite family of test functions.

The indicator weights $f(u) = \mathbf{1}_{[0,t]}(u)$ give the cycle-count process. The weights $f(u) = \sqrt{3}u\mathbf{1}_{[0,t]}(u)$ give the normalized additive order process. The least-common-multiple process then follows from a uniform additive-to-LCM transfer.

Proof architecture

The argument has five modules.

(i) The quantitative endpoint theory from the scalar paper gives a one-block bound for

$$\Delta_{n,r} = \mathbb{E}\,|S_{n,r}/n - 1|\,.$$

Repeating its dyadic summation without the logarithmic weight yields

$$\sum_j \sup_{2^j b_n \le r < 2^{j+1} b_n} \Delta_{n,r} = o(1).$$

(ii) The marked one-switch identity is tested against arbitrary bounded functions of the complete intermediate cycle vector. Away from the short source-cycle exception, the switch adds exactly one r-cycle and changes no other intermediate coordinate. This gives an approximate Campbell identity.

(iii) An immigration–death Stein equation converts the Campbell identity into total-variation approximation by a product of Poisson variables. The Stein constant is dimension-free.

(iv) The product-Poisson field satisfies a standard triangular-array martingale functional CLT on the logarithmic coordinate $u = \log r / \log n$. Total variation transfers this process theorem to the Luce cycle vector. Crude cycle bounds restore the omitted short and long ranges.

(v) The discrepancy $B_n(t) - \mathcal{O}_n(t)$ is nonnegative and nondecreasing in t. Hence its path supremum equals its endpoint value. The existing three-clock/von Mangoldt estimate

$$\mathbb{E}[B_n(1) - \mathcal{O}_n(1)] = O_\kappa(\log n \log\log n)$$

therefore transfers the whole additive process to the LCM process.

Status of this branch

The draft is written as a proof conditional on the same theorem-level inputs already used by the scalar paper. The only genuinely new probabilistic step is Theorem 1.3. The branch deliberately keeps the scalar characteristic-function proof and the Berry–Esseen supplement separate until the product-Poisson argument has been independently audited.

2 Model and imported switching inputs

After a common rescaling, assume throughout that

$$1 \le \theta_{n,i} \le \kappa, \qquad 1 \le i \le n, \tag{11}$$

for one fixed $\kappa < \infty$. Let

$$X_{n,i} \sim \mathrm{Exp}(\theta_{n,i}), \qquad 1 \le i \le n,$$

be independent, and let $\sigma_n(i)$ be the rank of $X_{n,i}$ among the n clocks. Write $C_{n,r}$ for the number of r-cycles and $L_n(i)$ for the cycle length containing i.

For distinct i, j, put

$$R_{ij} = \exp\{-(\theta_{n,i} - \theta_{n,j})(X_{n,j} - X_{n,i})\}, \qquad R_{ii} = 1, \tag{12}$$

and define

$$S_{n,r} = \sum_{i:L_n(i)>r} R_{i,\sigma_n^r(i)}. \tag{13}$$

Input theorem 2.1 (Marked one-switch identity, moments, and tails)**.** *For every $1 \le r < n$ and every bounded complex function $\Phi : S_n \to \mathbb{C}$,*

$$r(n-r)\mathbb{E}[C_{n,r}\Phi(\sigma_n)] = \mathbb{E}\sum_{i:L_n(i)>r} R_{i,\sigma_n^r(i)}\Phi\big(\sigma_n \circ (i\,\sigma_n^r(i))\big). \tag{14}$$

In particular,

$$r(n-r)\mathbb{E}C_{n,r} = \mathbb{E}S_{n,r}. \tag{15}$$

Put

$$\lambda_\kappa = \begin{cases} \infty, & \kappa = 1, \\ \kappa/(\kappa-1), & \kappa > 1. \end{cases}$$

For every $1 < p < \lambda_\kappa$,

$$\sup_{n,r} \frac{1}{n}\mathbb{E}\sum_{i:L_n(i)>r} R^p_{i,\sigma_n^r(i)} < \infty. \tag{16}$$

Moreover, the truncation and fresh-pair tail estimates stated in the scalar paper hold uniformly, and $\mathbb{E}R_{ij} = 1$ for every i, j.

This is the exact input used in the current scalar draft [9]. It implies the following crude bounds.

Lemma 2.2 (Crude cycle, root, and tail bounds)**.** *Uniformly in* $1 \le r < n$,

$$\mathbb{E}C_{n,r} \le \frac{C_\kappa n}{r(n-r)}. \tag{17}$$

If I_n *is uniform on* $[n]$*, independently of the clocks, then for* $a \le n/2$,

$$\mathbb{P}\{L_n(I_n) \le a\} \le C_\kappa a/n. \tag{18}$$

Consequently, for $b \le n/2$ *and* $U \le n/2$,

$$\mathbb{E}\sum_{r\le b} C_{n,r} = O_\kappa(\log(eb)), \tag{19}$$

$$\mathbb{E}\sum_{r>U} C_{n,r} = O_\kappa(1+\log(n/U)), \tag{20}$$

$$\mathbb{E}\sum_{r\le b} (\log r) C_{n,r} = O_\kappa((\log(eb))^2), \tag{21}$$

$$\mathbb{E}\sum_{r>U} (\log r) C_{n,r} \le C_\kappa \log n\{1+\log(n/U)\}. \tag{22}$$

Proof. Equation (17) follows from (15) and $\mathbb{E}S_{n,r} \le C_\kappa n$. Since

$$\mathbb{P}\{L_n(I_n) = r\} = \frac{r\mathbb{E}C_{n,r}}{n},$$

summing over $r \le a$ proves (18). For $r \le n/2$, (17) gives $\mathbb{E}C_{n,r} \le C_\kappa/r$. Harmonic summation proves the first and third estimates and the portions of the second and fourth estimates below $n/2$. There is deterministically at most one cycle longer than $n/2$. □

Fix once and for all

$$1 < p < \lambda_\kappa, \qquad \beta = 1 - \frac{1}{p} > 0. \tag{23}$$

The process argument needs a slightly stronger upper cutoff than the scalar CLT. To align this branch with the Berry–Esseen supplement, choose

$$A_0 > \max\left\{32\kappa^2, \frac{4}{\beta}, 4\right\}, \tag{24}$$

and put

$$b_n = \left\lceil (\log n)^3 \right\rceil, \qquad U_n = \left\lfloor \frac{n}{(\log n)^{A_0}} \right\rfloor. \tag{25}$$

For all sufficiently large n,

$$2U_n \le n/2. \tag{26}$$

Put

$$\Delta_{n,r} = \mathbb{E}\left|\frac{S_{n,r}}{n} - 1\right|. \tag{27}$$

For a dyadic lower endpoint a, set

$$\delta = \frac{4a}{n}, \qquad L_a = \log\frac{1}{\delta}, \qquad m = \left\lfloor \min\left\{\frac{a}{3}, \sqrt{L_a}\right\}\right\rfloor. \tag{28}$$

Input theorem 2.3 (Quantitative one-block switching)**.** *There is $C_\kappa < \infty$ such that, uniformly for $b_n \le a \le U_n$ and $a \le r < 2a$,*

$$\Delta_{n,r} \le C_\kappa(\Gamma_{n,a} + n^{-1})^\eta, \qquad \eta = \frac{1}{2\kappa}, \tag{29}$$

where

$$\Gamma_{n,a} \le C\Big[(1-\alpha_0)^{m-1} + m\delta^{1/4} + \frac{m^2}{n\delta^{1/2}} + m\delta^{1/(4\kappa)} + \frac{m\log n}{n\delta^{1/2}} + \delta^{1/(4\kappa)} + \delta^{1/2}L_a + \delta + g_{n,a}\Big], \tag{30}$$

with

$$g_{n,a} \le C\left[\delta^{-1/4}L_a e^{-cn\delta^{1/2}} + n^2\delta^{-1/4}e^{-cn\delta^{3/4}} + n^{-2}\right]. \tag{31}$$

All constants depend only on κ.

This theorem is the quantitative finite-endpoint interface proved and used in the current scalar paper. We also import the arithmetic output of its three-clock section.

Input theorem 2.4 (Two-cycle domination and endpoint product–LCM bound)**.** *Whenever $r + s \le n/2$,*

$$\mathbb{E}\left[C_{n,r}(C_{n,s} - \mathbf{1}_{\{r=s\}})\right] \le \frac{C_\kappa}{rs}. \tag{32}$$

Moreover, with

$$B_n = B_n(1) = \sum_{r=1}^{n}(\log r)C_{n,r}, \qquad O_n = \mathrm{ord}(\sigma_n),$$

one has

$$\mathbb{E}[B_n - \log O_n] \le C_\kappa \log n \log\log n. \tag{33}$$

The pair estimate follows from the exact rate-oriented three-clock identity; the endpoint discrepancy follows by the von Mangoldt overlap decomposition. Both are proved in [9].

3 Unweighted dyadic summability

The scalar proof sums $\Delta_{n,r}$ with a logarithmic block weight. The Campbell identity needs the stronger unweighted sum to tend to zero.

Let

$$a_j = 2^j b_n, \tag{34}$$

and sum over those j for which $a_j \le U_n$.

Proposition 3.1 (Unweighted dyadic switching summability)**.** *One has*

$$\boxed{\sum_{j:a_j \le U_n} \sup_{a_j \le r < 2a_j} \Delta_{n,r} \longrightarrow 0.} \tag{35}$$

The convergence is uniform over all rate arrays in $[1,\kappa]$.

Proof. Write

$$\delta_j = \frac{4a_j}{n}, \qquad L_j = \log \frac{1}{\delta_j}.$$

Since $a_j \geq b_n = (\log n)^3$ and $L_j \leq \log n$, the minimum in (28) is attained by $\sqrt{L_j}$ for all large n, uniformly in j. Thus $m_j \asymp \sqrt{L_j}$. Also

$$L_j \geq A_0 \log\log n - O(1). \tag{36}$$

Because $0 < \eta \leq 1$, the map $x \mapsto x^\eta$ is subadditive. We may therefore sum the contributions of the terms in (30) separately.

For the reset term,

$$\begin{aligned}\sum_j (1-\alpha_0)^{\eta(m_j-1)} &\leq C \sum_{L_j \geq A_0 \log\log n - O(1)} e^{-c\sqrt{L_j}} \\ &\leq C(1+\sqrt{\log\log n})e^{-c'\sqrt{\log\log n}} \longrightarrow 0.\end{aligned}$$

Here the L_j lie on an arithmetic lattice with fixed spacing.

After taking the η-power, every term involving a positive power of δ_j is bounded by

$$C L_j^d e^{-\gamma L_j}$$

for fixed $d \geq 0$ and $\gamma > 0$. The smallest exponent comes from $m\delta^{1/(4\kappa)}$ and is at least $1/(8\kappa^2)$. Hence

$$\sum_j L_j^d e^{-\gamma L_j} \leq C(\log\log n)^d (\log n)^{-\gamma A_0} \longrightarrow 0. \tag{37}$$

For the terms containing $(n\delta_j^{1/2})^{-1}$, use $m_j \leq \sqrt{\log n}$ and $a_j \geq (\log n)^3$ to obtain

$$\frac{m_j^2 + m_j \log n}{n\delta_j^{1/2}} \leq C \frac{(\log n)^{3/2}}{\sqrt{n a_j}} \leq \frac{C}{\sqrt{n}}.$$

After taking the η-power and summing over $O(\log n)$ blocks, the contribution is $O((\log n) n^{-\eta/2}) = o(1)$.

The first two terms in (31) are superpolynomially small uniformly for $a_j \geq (\log n)^3$. The n^{-2} term and the additional n^{-1} in (29) contribute $O((\log n)n^{-\eta}) = o(1)$. Combining the estimates proves (35). □

Remark 3.2 (A usable explicit error). The proof gives a deterministic $\varepsilon_n^{\mathrm{sw}} \downarrow 0$, depending only on κ, such that

$$\sum_j \sup_{a_j \leq r < 2a_j} \Delta_{n,r} \leq \varepsilon_n^{\mathrm{sw}}.$$

One may take a sum of terms of the schematic form

$$(1+\sqrt{\log\log n})e^{-c_\kappa\sqrt{\log\log n}}, \quad (\log\log n)^d (\log n)^{-A_0/(8\kappa^2)}, \quad (\log n) n^{-1/(4\kappa)}.$$

No optimization is needed for the qualitative process theorem.

4 An approximate Campbell identity

Let

$$\mathcal{I}_n = \{r \in \mathbb{Z} : b_n < r \le U_n\}, \qquad \xi_n = (C_{n,r})_{r\in\mathcal{I}_n} \in \mathbb{Z}_+^{\mathcal{I}_n}. \tag{38}$$

For $r \in \mathcal{I}_n$, let e_r denote the corresponding unit vector and put

$$a_{n,r} = \frac{n}{r(n-r)} = \frac{1}{r} + \frac{1}{n-r}. \tag{39}$$

The switch in (14) cuts a single source cycle at two marked points.

Lemma 4.1 (Exact intermediate-vector effect of a good split)**.** *Fix $r \in \mathcal{I}_n$ and a source label i with*

$$L_n(i) > r + U_n.$$

Then

$$\xi_n(\sigma_n \circ (i\, \sigma_n^r(i))) = \xi_n(\sigma_n) + e_r. \tag{40}$$

Proof. The transposition cuts the source cycle of length $L = L_n(i)$ into cycles of lengths r and $L - r$. The source length satisfies $L > U_n$, and the residual length satisfies $L - r > U_n$. Thus the source and residual lengths are both outside $\mathcal{I}_n$, while the new r-cycle lies in $\mathcal{I}_n$. All other cycles are unchanged. □

The exceptional source-cycle probability is uniformly summable.

Lemma 4.2 (Exceptional source-cycle estimate)**.** *Uniformly for $r \in \mathcal{I}_n$,*

$$\frac{1}{n}\mathbb{E} \sum_{i:L_n(i)>r} R_{i,\sigma_n^r(i)} \mathbf{1}_{\{L_n(i)\le r+U_n\}} \le C_\kappa \left(\frac{U_n}{n}\right)^\beta. \tag{41}$$

Proof. Let I_n be uniform on $[n]$, independently of the clocks. The left side is

$$\mathbb{E}\left[R_{I_n,\sigma_n^r(I_n)} \mathbf{1}_{\{r<L_n(I_n)\le r+U_n\}}\right].$$

Apply Hölder's inequality with exponents p and $p/(p-1)$. The selected pth moment is uniformly bounded by (16). Since $r + U_n \le 2U_n \le n/2$, (18) gives

$$\mathbb{P}\{L_n(I_n) \le r + U_n\} \le C_\kappa U_n/n.$$

The conjugate Hölder exponent contributes the power $\beta = 1 - 1/p$. □

Proposition 4.3 (Approximate Campbell identity)**.** *For every family of functions*

$$g_r : \mathbb{Z}_+^{\mathcal{I}_n} \to [-1, 1], \qquad r \in \mathcal{I}_n,$$

one has

$$\left| \mathbb{E} \sum_{r\in\mathcal{I}_n} C_{n,r} g_r(\xi_n - e_r) - \sum_{r\in\mathcal{I}_n} a_{n,r} \mathbb{E} g_r(\xi_n) \right| \le \varepsilon_n, \tag{42}$$

where

$$\varepsilon_n \le C_\kappa \varepsilon_n^{\mathrm{sw}} + C_\kappa (\log n) \left(\frac{U_n}{n}\right)^\beta \longrightarrow 0. \tag{43}$$

Here $g_r(\xi_n - e_r)$ is assigned any value when $C_{n,r} = 0$, since it is multiplied by $C_{n,r}$.

Proof. Fix $r \in \mathcal{I}_n$. Define a bounded function on permutations by

$$\Phi_r(\pi) = \begin{cases} g_r(\xi(\pi) - e_r), & C_r(\pi) \geq 1, \\ 0, & C_r(\pi) = 0, \end{cases}$$

where $\xi(\pi)$ is the intermediate cycle vector of π. Apply (14). Every switch on the right creates an r-cycle, so the first case in the definition of Φ_r applies there. On $\{L_n(i) > r + U_n\}$, Lemma 4.1 gives

$$\Phi_r\big(\sigma_n \circ (i\, \sigma_n^r(i))\big) = g_r(\xi_n).$$

On the complementary source-cycle range, the difference between this quantity and $g_r(\xi_n)$ is at most two. Dividing the marked identity by $r(n-r)$ therefore gives

$$\begin{aligned} &|\mathbb{E}[C_{n,r} g_r(\xi_n - e_r)] - a_{n,r}\mathbb{E} g_r(\xi_n)| \\ &\quad \leq a_{n,r}\Delta_{n,r} + 2a_{n,r}\frac{1}{n}\mathbb{E}\sum_{i: L_n(i) > r} R_{i,\sigma_n^r(i)} \mathbf{1}_{\{L_n(i) \leq r + U_n\}}. \end{aligned} \tag{44}$$

Since $r \leq U_n \leq n/2$, $a_{n,r} \leq 2/r$. On a dyadic block $a_j \leq r < 2a_j$,

$$\sum_{a_j \leq r < 2a_j} a_{n,r}\Delta_{n,r} \leq C \sup_{a_j \leq r < 2a_j} \Delta_{n,r}.$$

Proposition 3.1 handles the sum over blocks. By Lemma 4.2, the second contribution summed over r is at most

$$C_\kappa \left(\frac{U_n}{n}\right)^\beta \sum_{r \in \mathcal{I}_n} a_{n,r} \leq C_\kappa (\log n)^{1 - A_0\beta} = o(1),$$

because $A_0\beta > 1$. This proves (42). □

5 A dimension-free product-Poisson Stein lemma

We record the exact configuration-space statement needed to turn Proposition 4.3 into total variation.

Lemma 5.1 (Campbell-to-Poisson transfer)**.** *Let I be finite, let $X = (X_i)_{i \in I}$ take values in $\mathbb{Z}_+^I$, and let $\lambda_i \geq 0$. Suppose that for every family $g_i : \mathbb{Z}_+^I \to [-1, 1]$,*

$$\left| \mathbb{E}\sum_{i \in I} X_i g_i(X - e_i) - \sum_{i \in I} \lambda_i \mathbb{E} g_i(X) \right| \leq \varepsilon. \tag{45}$$

Then

$$d_{\mathrm{TV}}\left(\mathcal{L}(X), \bigotimes_{i \in I} \mathrm{Pois}(\lambda_i)\right) \leq \varepsilon. \tag{46}$$

The constant is independent of $|I|$ and of $\sum_i \lambda_i$.

Proof. Let $Z = (Z_i)_{i \in I}$ have the target product-Poisson law. Consider the immigration–death generator

$$\begin{aligned} \mathcal{A}h(\eta) = &\sum_{i \in I} \lambda_i [h(\eta + e_i) - h(\eta)] \\ &+ \sum_{i \in I} \eta_i [h(\eta - e_i) - h(\eta)]. \end{aligned} \tag{47}$$

For an indicator $f = \mathbf{1}_A$, let

$$h_f(\eta) = -\int_0^\infty \left(\mathbb{E}_\eta f(Y_t) - \mathbb{E} f(Z)\right)\, dt, \tag{48}$$

where (Y_t) is the immigration–death process with generator $\mathcal{A}$. Then

$$\mathcal{A} h_f = f - \mathbb{E} f(Z).$$

Couple processes started from η and $\eta + e_i$ by using the same immigrations and deaths. Their only discrepancy is the extra initial particle, which survives to time t with probability e^{-t}. Hence

$$\sup_{\eta, i} |h_f(\eta + e_i) - h_f(\eta)| \le \int_0^\infty e^{-t}\, dt = 1. \tag{49}$$

Put $g_i(\eta) = h_f(\eta + e_i) - h_f(\eta)$. Then

$$\begin{aligned} \mathbb{E} f(X) - \mathbb{E} f(Z) &= \mathbb{E} \mathcal{A} h_f(X) \\ &= \sum_i \lambda_i \mathbb{E} g_i(X) - \mathbb{E} \sum_i X_i g_i(X - e_i). \end{aligned}$$

Equation (45) bounds the absolute value by ε. Take the supremum over events A. □

Proof of Theorem 1.3. Apply Lemma 5.1 to Proposition 4.3 with $\lambda_r = a_{n,r}$. This gives

$$d_{\mathrm{TV}}\left(\mathcal{L}(\xi_n), \bigotimes_{r \in \mathcal{I}_n} \mathrm{Pois}(a_{n,r})\right) \le \varepsilon_n = o(1). \tag{50}$$

For Poisson variables,

$$d_{\mathrm{TV}}(\mathrm{Pois}(\lambda), \mathrm{Pois}(\mu)) \le |\lambda - \mu|.$$

Therefore

$$\begin{aligned} d_{\mathrm{TV}}\left(\bigotimes_{r \in \mathcal{I}_n} \mathrm{Pois}(a_{n,r}), \bigotimes_{r \in \mathcal{I}_n} \mathrm{Pois}(1/r)\right) &\le \sum_{r \in \mathcal{I}_n} \frac{1}{n - r} \\ &\le C \frac{U_n}{n} = o(1). \end{aligned}$$

The triangle inequality proves (8). □

Remark 5.2 (Why the theorem is genuinely high-dimensional). The number of coordinates is of order U_n, and the total Poisson intensity is of order $\log n$. The conclusion is not obtained by checking any fixed collection of cycle lengths. The dimension-free Stein factor in (49) is what permits the summation of the one-switch errors over the entire growing window.

6 The logarithmic Poisson field

We first prove the functional theorem for the independent Poisson reference process. Let $(Z_{n,r})_{r \in \mathcal{I}_n}$ be independent with

$$Z_{n,r} \sim \mathrm{Pois}(a_{n,r}), \tag{51}$$

and put

$$\ell_n = \log n, \qquad u_{n,r} = \frac{\log r}{\ell_n}.$$

Define the two-dimensional process

$$M_n(t) = \frac{1}{\sqrt{\ell_n}} \sum_{\substack{r \in \mathcal{I}_n \\ u_{n,r} \le t}} (Z_{n,r} - a_{n,r}) \begin{pmatrix} 1 \\ \sqrt{3}\, u_{n,r} \end{pmatrix}, \qquad 0 \le t \le 1. \tag{52}$$

6.1 Uniform logarithmic Riemann sums

Lemma 6.1 (Uniform logarithmic moment sums)**.** *For $k = 0, 1, 2$,*

$$\sup_{0 \le t \le 1} \left| \frac{1}{\ell_n^{k+1}} \sum_{\substack{r \in \mathcal{I}_n \\ r \le n^t}} a_{n,r} (\log r)^k - \frac{t^{k+1}}{k+1} \right| \longrightarrow 0. \tag{53}$$

Proof. Use

$$a_{n,r} = \frac{1}{r} + \frac{1}{n-r}.$$

The $1/(n-r)$ contribution is bounded uniformly in t by

$$\frac{C}{\ell_n^{k+1}} \frac{U_n}{n} (\log n)^k = o(1).$$

For the $1/r$ contribution, integral comparison gives, uniformly for $2 \le x \le n$,

$$\sum_{r \le x} \frac{(\log r)^k}{r} = \frac{(\log x)^{k+1}}{k+1} + O_k(1 + (\log x)^k / x).$$

Restricting to $b_n < r \le U_n$ removes at most

$$O((\log b_n)^{k+1}) + O(\ell_n^k \log(n/U_n)) = O(\ell_n^k \log \ell_n).$$

After division by ℓ_n^{k+1} this is $o(1)$, uniformly in t. □

The deterministic covariance matrix of $M_n(t)$ is

$$Q_n(t) = \frac{1}{\ell_n} \sum_{\substack{r \in \mathcal{I}_n \\ u_{n,r} \le t}} a_{n,r} \begin{pmatrix} 1 & \sqrt{3} u_{n,r} \\ \sqrt{3} u_{n,r} & 3 u_{n,r}^2 \end{pmatrix}. \tag{54}$$

By Lemma 6.1,

$$\sup_{0 \le t \le 1} \| Q_n(t) - Q(t) \| \longrightarrow 0, \qquad Q(t) = \begin{pmatrix} t & \frac{\sqrt{3}}{2} t^2 \\ \frac{\sqrt{3}}{2} t^2 & t^3 \end{pmatrix}. \tag{55}$$

6.2 A triangular-array functional CLT

Proposition 6.2 (Poisson functional CLT)**.** *The processes M_n converge in $D([0,1],\mathbb{R}^2)$ to*

$$G(t) = \left(W(t), \sqrt{3}\int_0^t u\, dW(u)\right). \tag{56}$$

Proof. Order the independent centered increments in (52) by increasing r. Their covariance partial sums converge uniformly to $Q(t)$ by (55). It remains to verify the Lindeberg condition.

Let

$$v_{n,r} = \frac{1}{\sqrt{\ell_n}}\begin{pmatrix} 1 \\ \sqrt{3}u_{n,r}\end{pmatrix}.$$

Then $\|v_{n,r}\| \le 2/\sqrt{\ell_n}$. If $Z \sim \mathrm{Pois}(\lambda)$, then

$$\mathbb{E}(Z-\lambda)^4 = \lambda + 3\lambda^2.$$

Consequently,

$$\begin{aligned}\sum_{r\in\mathcal{I}_n} \mathbb{E}\,\|v_{n,r}(Z_{n,r}-a_{n,r})\|^4 &\le \frac{C}{\ell_n^2}\sum_{r\in\mathcal{I}_n}(a_{n,r}+a_{n,r}^2)\\ &= O(\ell_n^{-1}).\end{aligned}$$

For every $\varepsilon > 0$, the Lindeberg sum is bounded by ε^{-2} times this fourth-moment sum and therefore tends to zero. The standard martingale or independent-triangular-array functional CLT now gives a continuous centered Gaussian martingale with covariance Q. Since

$$Q(t) = \int_0^t \begin{pmatrix} 1 \\ \sqrt{3}u\end{pmatrix}\begin{pmatrix} 1 & \sqrt{3}u\end{pmatrix} du,$$

the limit has the representation (56). □

Remark 6.3 (Continuous-limit tightness). The fourth-moment calculation also implies that the maximal jump tends to zero in probability. Hence the sequence is C-tight, although we state the result in the standard J_1 topology on $D([0,1],\mathbb{R}^2)$.

7 Transfer to the Luce intermediate field

Define

$$\widehat{M}_n(t) = \frac{1}{\sqrt{\ell_n}}\sum_{\substack{r\in\mathcal{I}_n\\ u_{n,r}\le t}}(C_{n,r}-a_{n,r})\begin{pmatrix} 1 \\ \sqrt{3}u_{n,r}\end{pmatrix}. \tag{57}$$

Proposition 7.1 (Intermediate Luce functional CLT)**.** *One has*

$$\widehat{M}_n \Rightarrow \left(W(t), \sqrt{3}\int_0^t u\, dW(u)\right) \tag{58}$$

in $D([0,1],\mathbb{R}^2)$.

Proof. The map from the finite vector $(x_r)_{r\in\mathcal{I}_n}$ to the path in (57) is measurable. Total variation cannot increase under a measurable map. Theorem 1.3 therefore gives

$$d_{\mathrm{TV}}(\mathcal{L}(\widehat{M}_n), \mathcal{L}(M_n)) \longrightarrow 0.$$

Apply Proposition 6.2. □

The same argument gives arbitrary finite families of bounded logarithmic weights.

Proposition 7.2 (Intermediate Gaussian field)**.** *Let $f_1, \dots, f_d$ be bounded Riemann-integrable functions on $[0,1]$. Then*

$$\left(\frac{1}{\sqrt{\ell_n}} \sum_{r\in\mathcal{I}_n} f_j(u_{n,r})(C_{n,r} - a_{n,r})\right)_{j=1}^{d} \Rightarrow N(0,\Sigma), \qquad \Sigma_{ij} = \int_0^1 f_i(u) f_j(u)\, du. \tag{59}$$

Proof. Transfer to the product-Poisson field by total variation. For any fixed linear combination $f = \sum_j t_j f_j$, the centered Poisson sum has variance

$$\frac{1}{\ell_n} \sum_{r\in\mathcal{I}_n} a_{n,r} f(u_{n,r})^2 \longrightarrow \int_0^1 f(u)^2\, du.$$

The maximal coefficient is $O(\ell_n^{-1/2})$, and the same fourth-moment argument as above gives Lindeberg. Apply Cramér–Wold. □

8 Restoring the full cycle processes

Define the intermediate count and additive processes

$$K_n^\circ(t) = \sum_{\substack{r\in\mathcal{I}_n \\ r\le n^t}} C_{n,r}, \tag{60}$$

$$B_n^\circ(t) = \sum_{\substack{r\in\mathcal{I}_n \\ r\le n^t}} (\log r) C_{n,r}. \tag{61}$$

Because all omitted summands are nonnegative,

$$\sup_{0\le t\le 1} |K_n(t) - K_n^\circ(t)| \le \sum_{r\le b_n} C_{n,r} + \sum_{r>U_n} C_{n,r}, \tag{62}$$

$$\sup_{0\le t\le 1} |B_n(t) - B_n^\circ(t)| \le \sum_{r\le b_n} (\log r) C_{n,r} + \sum_{r>U_n} (\log r) C_{n,r}. \tag{63}$$

Lemma 8.1 (Uniform restoration of omitted cycle lengths)**.** *One has*

$$\sup_{0\le t\le 1} \frac{|K_n(t) - K_n^\circ(t)|}{\sqrt{\ell_n}} \longrightarrow 0 \qquad \text{in } L^1, \tag{64}$$

$$\sup_{0\le t\le 1} \frac{|B_n(t) - B_n^\circ(t)|}{\ell_n^{3/2}} \longrightarrow 0 \qquad \text{in } L^1. \tag{65}$$

Proof. By Lemma 2.2, the expectation of the right side in (62) is

$$O_\kappa(\log b_n + 1 + \log(n/U_n)) = O_\kappa(\log \ell_n) = o(\sqrt{\ell_n}).$$

The expectation of the right side in (63) is

$$O_\kappa((\log b_n)^2 + \ell_n\{1 + \log(n/U_n)\}) = O_\kappa(\ell_n \log \ell_n) = o(\ell_n^{3/2}).$$

□

We next replace the exact intermediate Poisson centers by the classical centers.

Lemma 8.2 (Uniform deterministic centering)**.** *Uniformly for* $0 \le t \le 1$,

$$\sum_{\substack{r\in\mathcal{I}_n\\ r\le n^t}} a_{n,r} = t\ell_n + O(\log \ell_n), \tag{66}$$

$$\sum_{\substack{r\in\mathcal{I}_n\\ r\le n^t}} a_{n,r}\log r = \frac{1}{2}t^2\ell_n^2 + O(\ell_n \log \ell_n). \tag{67}$$

Proof. This is Lemma 6.1 for $k = 0, 1$, with the error from deleting the lower and upper ranges retained rather than divided by the leading scale. The $1/(n-r)$ portion contributes $O(U_n/n)$ in (66) and $O(\ell_n U_n/n)$ in (67). □

Theorem 8.3 (Bivariate additive functional theorem)**.** *In* $D([0,1],\mathbb{R}^2)$,

$$\boxed{\left(\frac{K_n(t) - t\ell_n}{\sqrt{\ell_n}}, \frac{B_n(t) - \frac{1}{2}t^2\ell_n^2}{\sqrt{\ell_n^3/3}}\right)_{0\le t\le 1} \Rightarrow \left(W(t), \sqrt{3}\int_0^t u\, dW(u)\right)_{0\le t\le 1}.} \tag{68}$$

Proof. The two coordinates of (57) are

$$\frac{K_n^\circ(t) - \sum_{r\in\mathcal{I}_n, r\le n^t} a_{n,r}}{\sqrt{\ell_n}}$$

and

$$\frac{B_n^\circ(t) - \sum_{r\in\mathcal{I}_n, r\le n^t} a_{n,r}\log r}{\sqrt{\ell_n^3/3}}.$$

Apply Proposition 7.1, then use Lemmas 8.1 and 8.2 and Slutsky's theorem in $D([0,1],\mathbb{R}^2)$. □

Proof of Theorem 1.4. Apply Proposition 7.2. Replacing $a_{n,r}$ by $1/r$ changes each normalized deterministic center by

$$O\left(\frac{1}{\sqrt{\ell_n}}\sum_{r\le U_n}\frac{1}{n-r}\right) = o(1).$$

For a bounded f, the omitted random contribution is bounded by $\|f\|_\infty$ times the omitted cycle count in (62), which is $o_{L^1}(\sqrt{\ell_n})$. The omitted $1/r$ center has size $O(\log \ell_n)$. Cramér–Wold completes the proof. □

9 Functional additive-to-LCM transfer

For $m \geq 1$, define the cumulative divisibility count

$$D_{n,m}(t) = \sum_{\substack{r \leq n^t \\ m \mid r}} C_{n,r}. \tag{69}$$

Let Λ be the von Mangoldt function. The identities

$$\log r = \sum_{m \mid r} \Lambda(m)$$

and

$$\mathcal{O}_n(t) = \sum_{m \leq n^t} \Lambda(m) \mathbf{1}_{\{D_{n,m}(t) \geq 1\}}$$

give the exact process decomposition

$$B_n(t) - \mathcal{O}_n(t) = \sum_{m \leq n^t} \Lambda(m)(D_{n,m}(t) - 1)_+. \tag{70}$$

Lemma 9.1 (Monotonicity of the product–LCM discrepancy)**.** *The process*

$$t \longmapsto B_n(t) - \mathcal{O}_n(t)$$

is nonnegative and nondecreasing. Consequently,

$$\sup_{0 \leq t \leq 1} |B_n(t) - \mathcal{O}_n(t)| = B_n(1) - \log \operatorname{ord}(\sigma_n). \tag{71}$$

Proof. For every m, $D_{n,m}(t)$ is nondecreasing in t, and $x \mapsto (x-1)_+$ is nondecreasing. Apply (70). □

Proposition 9.2 (Uniform functional LCM transfer)**.** *One has*

$$\sup_{0 \leq t \leq 1} \frac{|B_n(t) - \mathcal{O}_n(t)|}{\ell_n^{3/2}} \longrightarrow 0 \tag{72}$$

in L^1.

Proof. By Lemma 9.1 and Input Theorem 2.4,

$$\mathbb{E} \sup_{0 \leq t \leq 1} |B_n(t) - \mathcal{O}_n(t)| = \mathbb{E}[B_n - \log O_n] \leq C_\kappa \ell_n \log \ell_n = o(\ell_n^{3/2}).$$

□

Proof of Theorem 1.1. After a common rescaling, the rate hypothesis is exactly (11). Theorem 8.3 gives the bivariate count/additive-process limit. Proposition 9.2 replaces $B_n(t)$ by $\mathcal{O}_n(t)$ uniformly in t at the second-coordinate scale. Slutsky's theorem proves (3). □

10 Multivariate consequences

The process theorem immediately gives all fixed finite-time joint laws.

Corollary 10.1 (Finite-time multivariate law)**.** *Fix $0 \le t_1 < \cdots < t_d \le 1$. The $2d$-vector consisting of*

$$\frac{K_n(t_j) - t_j \ell_n}{\sqrt{\ell_n}} \quad \text{and} \quad \frac{\mathcal{O}_n(t_j) - \frac{1}{2} t_j^2 \ell_n^2}{\sqrt{\ell_n^3/3}}, \qquad 1 \le j \le d,$$

converges to a centered Gaussian vector whose covariance is determined by (4)–(6).

Proof. Apply the continuous evaluation map at the fixed times to Theorem 1.1; the limit paths are continuous. □

Proof of Corollary 1.2. Take $t = 1$ in Theorem 1.1. Equations (4)–(6) give the displayed covariance matrix. □

Corollary 10.2 (Independent logarithmic annuli)**.** *Fix disjoint intervals $(s_j, t_j] \subset [0,1]$. The corresponding normalized increment vectors*

$$\left(\frac{K_n(t_j) - K_n(s_j) - (t_j - s_j)\ell_n}{\sqrt{\ell_n}}, \frac{\mathcal{O}_n(t_j) - \mathcal{O}_n(s_j) - \frac{1}{2}(t_j^2 - s_j^2)\ell_n^2}{\sqrt{\ell_n^3/3}} \right)$$

converge jointly to independent centered Gaussian vectors. For one annulus $(s,t]$, the covariance matrix is

$$\begin{pmatrix} t - s & \frac{\sqrt{3}}{2}(t^2 - s^2) \\ \frac{\sqrt{3}}{2}(t^2 - s^2) & t^3 - s^3 \end{pmatrix}.$$

Proof. The limiting increments are stochastic integrals against W over disjoint intervals and are therefore independent. □

Corollary 10.3 (Time-changed Brownian order process)**.** *The second coordinate of Theorem 1.1 may be written as*

$$\big(B(t^3)\big)_{0 \le t \le 1}$$

in distribution, where B is standard Brownian motion.

Proof. The process $\sqrt{3}\int_0^t u \, dW(u)$ is a centered continuous Gaussian martingale with quadratic variation

$$3\int_0^t u^2 \, du = t^3.$$

Apply the Dambis–Dubins–Schwarz representation, or compare Gaussian covariances directly. □

At the endpoint, the common-noise representation gives a useful regression identity.

Corollary 10.4 (Endpoint Gaussian regression)**.** *Let (Z_K, Z_O) denote the limiting endpoint pair. Then*

$$Z_O \overset{d}{=} \frac{\sqrt{3}}{2} Z_K + \frac{1}{2} Z_\perp,$$

where $Z_K, Z_\perp$ are independent standard normals. Thus the normalized cycle count explains three quarters of the limiting order variance.

11 Relation to the Berry–Esseen branch

The product-Poisson theorem is a qualitative process tool. It should not be confused with the frequency-sensitive comparison used in the quantitative supplement [8].

The total-variation error in (43) contains the reset term

$$\exp\{-c_\kappa \sqrt{\log\log n}\},$$

up to polylogarithmic factors. This tends to zero, but it is much larger than the natural scalar Berry–Esseen scale $(\log n)^{-1/2}$. In the scalar characteristic-function ODE, the same raw switching error is divided by the standard deviation and integrated over a controlled frequency window. The Berry–Esseen supplement exploits that extra structure to obtain the sharp $O_\kappa((\log n)^{-1/2})$ rate for the additive proxy.

Accordingly, the recommended division of labor is:

1. Use Theorem 1.3 and the present branch for qualitative functional and multivariate universality.
2. Retain the scalar switching ODE for sharp additive Berry–Esseen and Edgeworth analysis.
3. Use the multi-clock moment bounds from the quantitative supplement for concentration of the arithmetic remainder and order-level rates.
4. Pursue a quantitative functional theorem only after selecting an explicit path metric. A sharp endpoint Kolmogorov rate and a path-space bounded-Lipschitz rate are different problems and need not have the same proof.

Remark 11.1 (Why this branch still helps quantitative work). The product-Poisson approximation identifies the correct Gaussian covariance operator and proves that no hidden profile-dependent covariance survives on logarithmic scale. It therefore supplies the structural target for any future multivariate Stein, strong approximation, or functional Berry–Esseen theorem, even though its present total-variation error is not sharp enough for the scalar rate.

12 Audit checklist and merge plan

Before this branch replaces the scalar section in the main paper, the following points should be checked independently.

(i) Re-run Proposition 3.1 directly from the exact constants in the finite-endpoint input, with the stronger choice $A_0\beta > 1$ recorded explicitly.

(ii) Verify the permutation convention in Lemma 4.1 against the marked switch used in the companion manuscript: right-composition by $(i\,\sigma_n^r(i))$ must split the source cycle into lengths r and $L-r$ with no orientation reversal.

(iii) Check the factor two and normalization $a_{n,r} = n/[r(n-r)]$ in (44).

(iv) Keep the Stein lemma in the paper, rather than relying on an unstated high-dimensional Poisson approximation theorem. Its dimension-free first difference bound is the central reason the argument closes.

(v) Verify the uniform-in-t deterministic center estimates at the two cutoff transition times $t = \log b_n / \log n$ and $t = \log U_n / \log n$.

(vi) Preserve the scalar characteristic-function proof in a separate branch until the product-Poisson theorem has passed this audit. Afterward, either move the scalar proof to an appendix or retain it only in the Berry–Esseen supplement.

A merged final paper would naturally have the following structure:

1. Introduction and the bivariate functional theorem.
2. Model, marked one-switch input, and quantitative endpoint interface.
3. Unweighted dyadic summability.
4. Product-Poisson approximation of the intermediate cycle field.
5. Gaussian random measure and functional additive laws.
6. Rate-oriented three-clock switch and factorial moments.
7. Functional product–LCM transfer.
8. Scalar, joint, and annular consequences.
9. Quantitative directions and relation to the Berry–Esseen supplement.

A A general weighted additive-to-LCM process transfer

The monotonicity argument is independent of permutations and may be useful in other component models.

Let $(C_{n,r})_{r\ge 1}$ be arbitrary nonnegative integer-valued variables and let $x_n(t)$ be a nondecreasing integer-valued cutoff. Define

$$A_n(t) = \sum_{r \le x_n(t)} C_{n,r} \log r,$$

and

$$L_n(t) = \log \operatorname{lcm}\{r \le x_n(t) : C_{n,r} > 0\}.$$

Proposition A.1 (Abstract functional additive-to-LCM transfer)**.** *The discrepancy $A_n(t) - L_n(t)$ is nonnegative and nondecreasing. Hence, for any deterministic scale $s_n > 0$,*

$$\sup_t \frac{|A_n(t) - L_n(t)|}{s_n} = \frac{A_n(1) - L_n(1)}{s_n}.$$

Consequently, if

$$\frac{A_n(\cdot) - a_n(\cdot)}{s_n} \Rightarrow X(\cdot)$$

in D and

$$\frac{A_n(1) - L_n(1)}{s_n} \longrightarrow 0$$

in probability, then

$$\frac{L_n(\cdot) - a_n(\cdot)}{s_n} \Rightarrow X(\cdot).$$

Proof. Put

$$D_{n,m}(t) = \sum_{\substack{r \le x_n(t) \\ m \mid r}} C_{n,r}.$$

Then

$$A_n(t) - L_n(t) = \sum_m \Lambda(m)(D_{n,m}(t) - 1)_+.$$

Each summand is nonnegative and nondecreasing in t. The functional transfer is Slutsky's theorem under the uniform perturbation displayed above. □

B A finite-dimensional covariance formula

For later quantitative work, it is useful to record the covariance of all linear combinations of the two limiting processes. Let

$$X(t) = W(t), \qquad Y(t) = \sqrt{3} \int_0^t u \, dW(u).$$

For coefficients a_j, b_j and times t_j,

$$\sum_j \{a_j X(t_j) + b_j Y(t_j)\} = \int_0^1 \left[\sum_j \{a_j + \sqrt{3} b_j u\} \mathbf{1}_{\{u \le t_j\}} \right] dW(u).$$

Therefore its variance is

$$\int_0^1 \left[\sum_j \{a_j + \sqrt{3} b_j u\} \mathbf{1}_{\{u \le t_j\}} \right]^2 du. \tag{73}$$

This is the covariance form that should be preserved in any multivariate Berry–Esseen or Edgeworth extension.

References

[1] R. Arratia, A. D. Barbour, and S. Tavaré, *Logarithmic Combinatorial Structures: A Probabilistic Approach*, EMS Monographs in Mathematics, European Mathematical Society, Zürich, 2003.

[2] A. D. Barbour, L. Holst, and S. Janson, *Poisson Approximation*, Oxford Studies in Probability, vol. 2, Clarendon Press, Oxford, 1992.

[3] A. D. Barbour and S. Tavaré, A rate for the Erdős–Turán law, *Combinatorics, Probability and Computing* 3 (1994), no. 2, 167–176.

[4] P. Billingsley, *Convergence of Probability Measures*, second edition, Wiley Series in Probability and Statistics, Wiley, New York, 1999.

[5] S. Chatterjee, P. Diaconis, and G. B. Kim, Enumerative theory for the Tsetlin library, *Journal of Algebra* 655 (2024), 139–162.

[6] P. Erdős and P. Turán, On some problems of a statistical group theory. III, *Acta Mathematica Academiae Scientiarum Hungaricae* 18 (1967), 309–320.

[7] S. N. Ethier and T. G. Kurtz, *Markov Processes: Characterization and Convergence*, Wiley Series in Probability and Mathematical Statistics, Wiley, New York, 1986.

[8] C. D. Long, Berry–Esseen checkpoints for the Erdős–Turán law of comparable-rate Luce permutations, research supplement, 2026.

[9] C. D. Long, An Erdős–Turán law for comparable-rate Luce permutations, manuscript, 2026.

[10] C. D. Long, Mesoscopic cycles of comparable-rate Luce permutations: a switching proof under submacroscopic root-cycle escape, manuscript, 2026.

[11] R. D. Luce, *Individual Choice Behavior: A Theoretical Analysis*, John Wiley & Sons, New York, 1959.